Morphology and word-formation

*Introduction to linguistics — **Course Website** — 1 & 8 November 2025*

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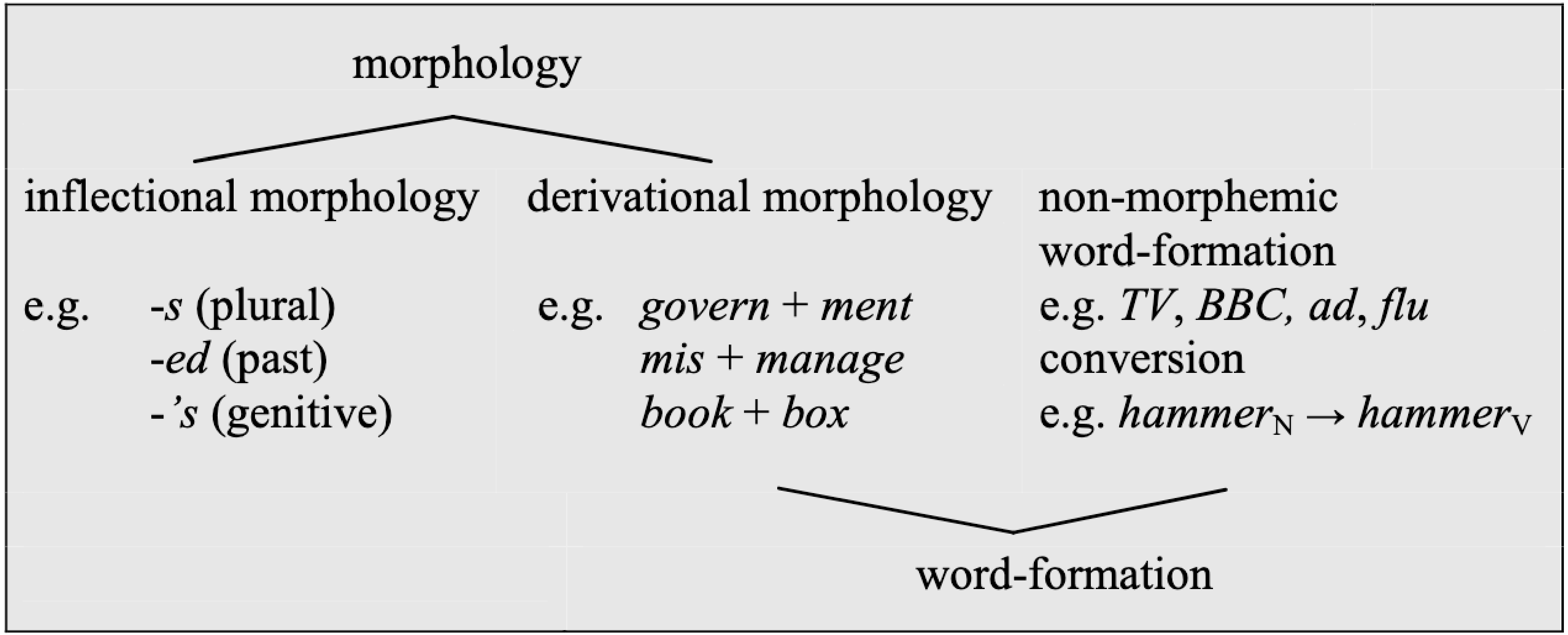
Morphology

Morphology and word-formation



Schmid (2016): 15

Object of study

“Morphology is concerned with the internal structure of words and with the various processes which allow us to constantly expand the vocabulary of a language.” (Kortmann 2020: 51)

Morphemes

"**Morpheme**: The basic morphological unit is not the word, but the morpheme (Greek morphé = shape, form), the smallest meaning-bearing unit of language.

Thus, the word *singers* contains three morphemes: {sing}, {-er}, and {-s}. Each of these three morphemes adds to the overall meaning of singers:

- the verb {sing} makes the central contribution,
- while {-er} on its own means no more than 'someone who VERBs',
- and the {-s} merely gives grammatical information, namely plural."

(Kortmann 2020: 51)

Inflectional morphology

word class	kind of inflection	inflectional morphemes	examples	number of word forms
noun	declension	{PLURAL}: {-s}	two boy-s	rule: 2
		{'GENITIVE'}: {-s}	the boy-'s toy	exception: 4
verb	conjugation	{3SG.IND.PRES}:{-s}	he work-s	rule: 4
		{PAST}:{-ed}	he work-ed	exception: 5 (8)
		{PRES.PART.}:{-ing}	he is work-ing	
		{PAST PART.}:{-ed}	he has work-ed	
adjective	comparison	{COMPARATIVE}:{-er}	strong-er	rule: 3
		{SUPERLATIVE}:{-est}	strong-est	

Kortmann (2020): 81

Allomorphs

- “**Morphemes vs (allo-)morphs**: On the level of morphology, morphemes are the exact **counterpart to phonemes** on the level of phonology.
- Just like phonemes, morphemes are abstract units which can be realized by more than one form.
- Just think of the plural morpheme in word forms like *kids*, *kits*, and *kisses*, where it is realized as /-z/, /-s/, and /-ɪz/ respectively.
- These **concrete realizations of morphemes are called morphs**, and in analogy to the allophones of a phoneme we speak of the allomorphs of a morpheme”

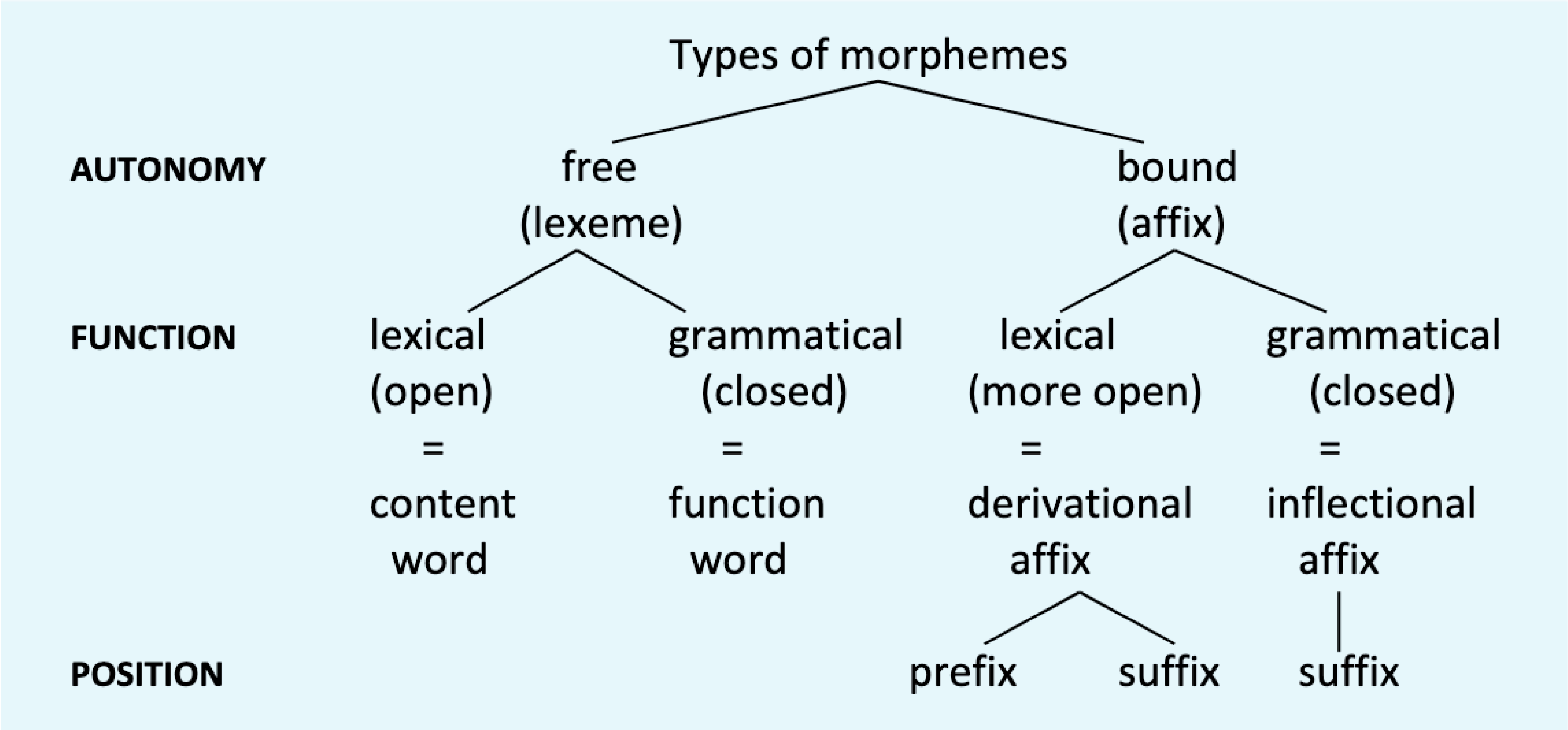
(Kortmann 2020: 51)

Word, word form, lexeme

- "**Word – word form – lexeme**: The example *singers* also serves to show how important it is to deal with the term word in a more differentiated way.
- Is *singers* a different **word** from *singer*? No, singers is merely a different form, a so-called **word form**, of the noun *singer*.
- *Singer* itself, on the other hand, is not merely a word form of *sing*, but a different, new word, which has been formed by affixation of {-er} to the verb.
- In this case, we speak of a new **lexeme** with a new dictionary entry which has been created by a specific derivational process."

(Kortmann 2020: 51)

Types of morphemes



Kortmann (2020): 53

morpheme class	attributes	examples
lexically free (lexemes, roots)	• semantically autonomous • open class • inflecting (except for interjections)	N: {head}, {key}, {part} Adj: {green}, {pale}, {nice} Adv: {fast}, {well}, {here} V lex: {plan}, {teach}, {pass} Interj: {ouch}, {damn}
grammatically free (lexemes, function words)	• not autonomous, relational • closed class • not inflecting	V aux: {will}, {must}, {have} Pron: {she}, {which}, {why} Prep: {of}, {in}, {at} Conj: {and}, {that}, {as} Num: {one}, {thousand}
lexically bound (derivational affixes)	• (more or less) open class • create lexemes with discrete meaning • usually change word class (suffixes) • closer to the stem	prefixes: {re-}, {dis-}, {post-} suffixes: {-er}, {-ify}, {-ment}
grammatically bound (inflectional suffixes)	• closed class • create word-forms with unchanged meaning • always maintain word class • further away from the stem	for Ns: {plural}, {genitive} for Vs: {3^{rd} pers. pres. sg.}, {ing}, {ed_1}, {ed_2} for Adjs: {er}, {est} for Nums: {th}

Schmid (2016): 36

Exercise: determine morpheme types

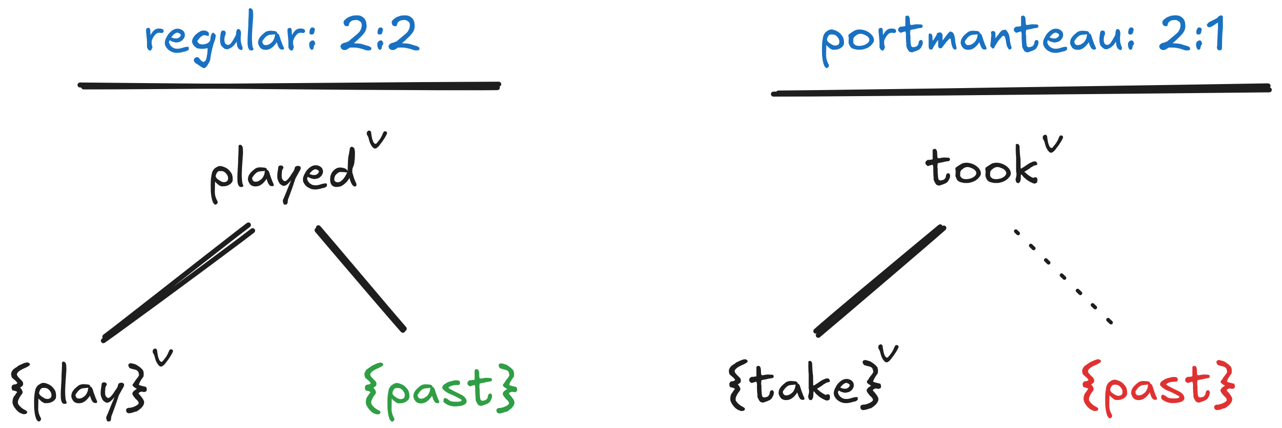
Divide the words below into their component morphemes and classify them.

1. *signpost*: $\{\text{sign}\}_{\text{lx.fr}}$ $\{\text{post}\}_{\text{lx.fr}}$
2. *cats*: $\{\text{cat}\}_{\text{lx.bd}}$ $\{\text{plural}\}_{\text{gr.bd}}$
3. *they*: $\{\text{they}\}_{\text{gr.fr}}$
4. *singer*: $\{\text{sing}\}_{\text{lx.fr}}$ $\{\text{er}\}_{\text{lx.bd}}$
5. *finger*: $\{\text{finger}\}_{\text{lx.fr}}$
6. *funnier*: $\{\text{funny}\}_{\text{lx.fr}}$ $\{\text{er}\}_{\text{gr.bd}}$
7. *undeniable*: $\{\text{un}\}_{\text{lx.bd}}$ $\{\text{deny}\}_{\text{lx.fr}}$ $\{\text{able}\}_{\text{lx.bd}}$

Special types of morphemes

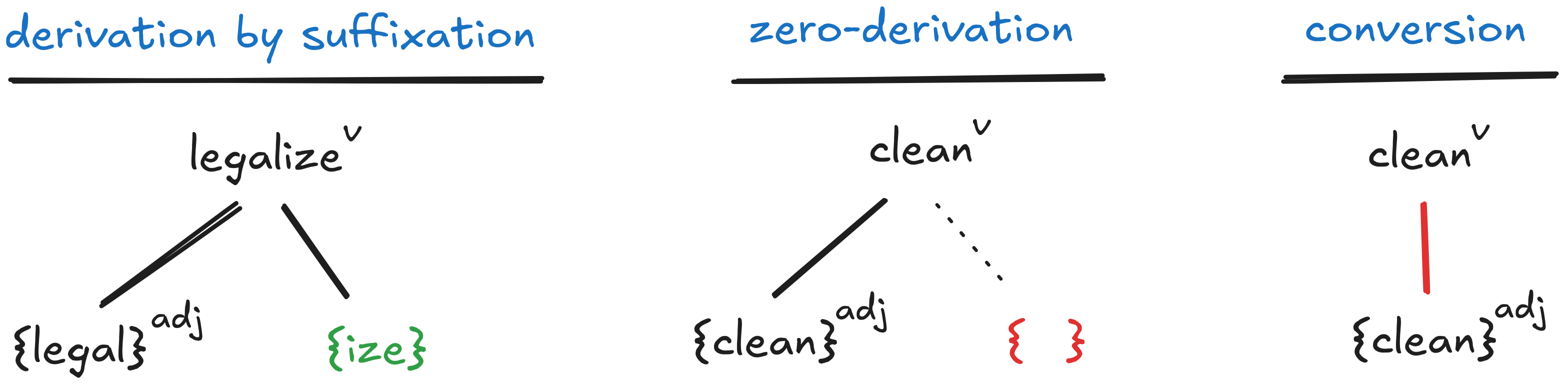
Portmanteau morphemes

- morphemes that carries several meanings simultaneously
- i.e. that carry information which is usually expressed by way of two or more morphemes
- e.g. *took*, *taught*



Zero morphemes

- an inflectional or word-formation morpheme lacking phonetic substance
- e.g. {plural} morpheme in *two sheep*

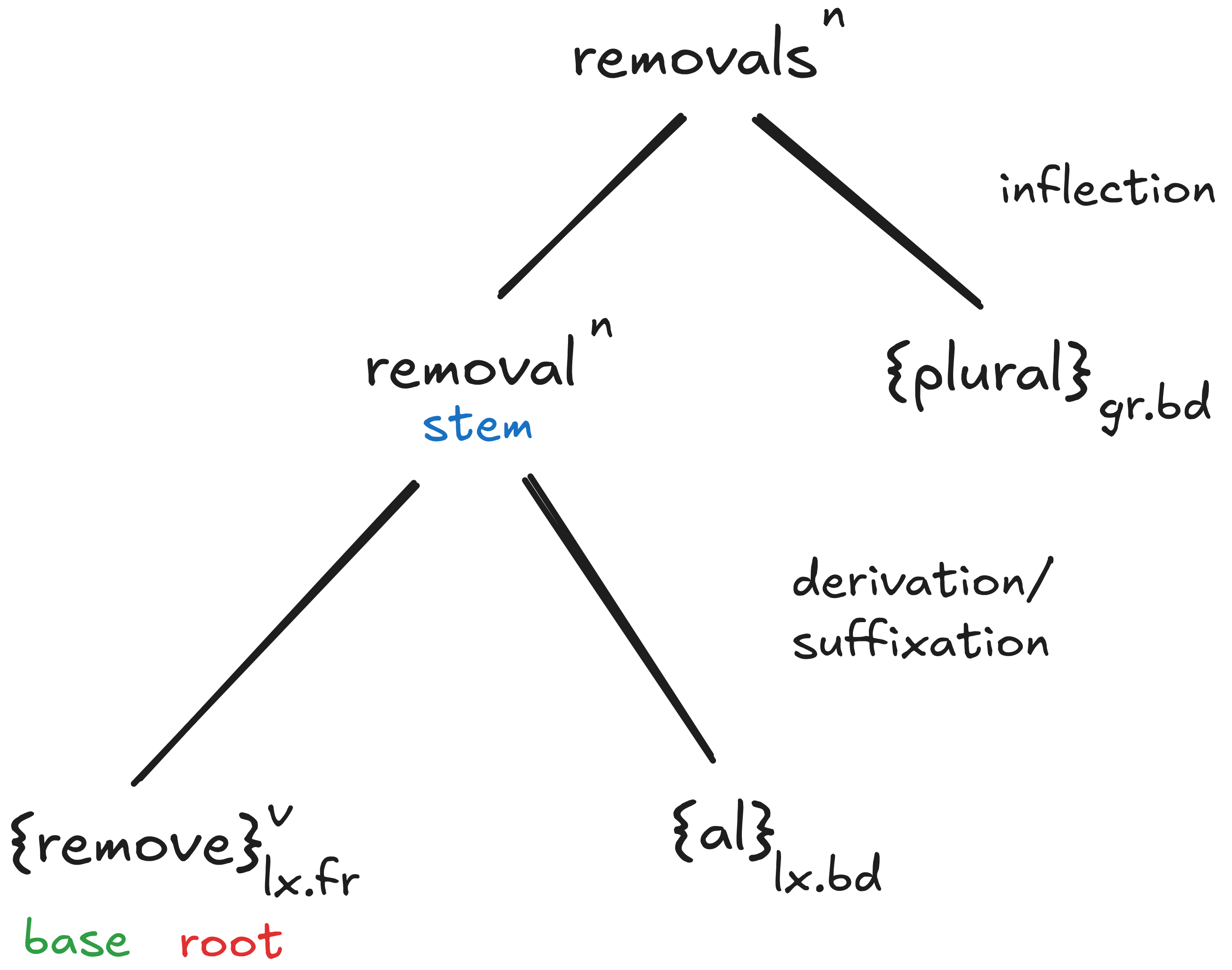


Base, root, stem

Example: *removals*

- "What remains once all inflectional suffixes (here plural {-s}) are taken away is the **stem**.
- What remains when taking away all affixes ({-re}, {-al} and {-s}) is the **root**, which is the minimal lexical unit and cannot be morphologically analysed any further.
- What remains if individual derivational affixes are taken away from the stem (here: {-al} from removal) is called the **base** in the narrower sense - which is still larger than the root."

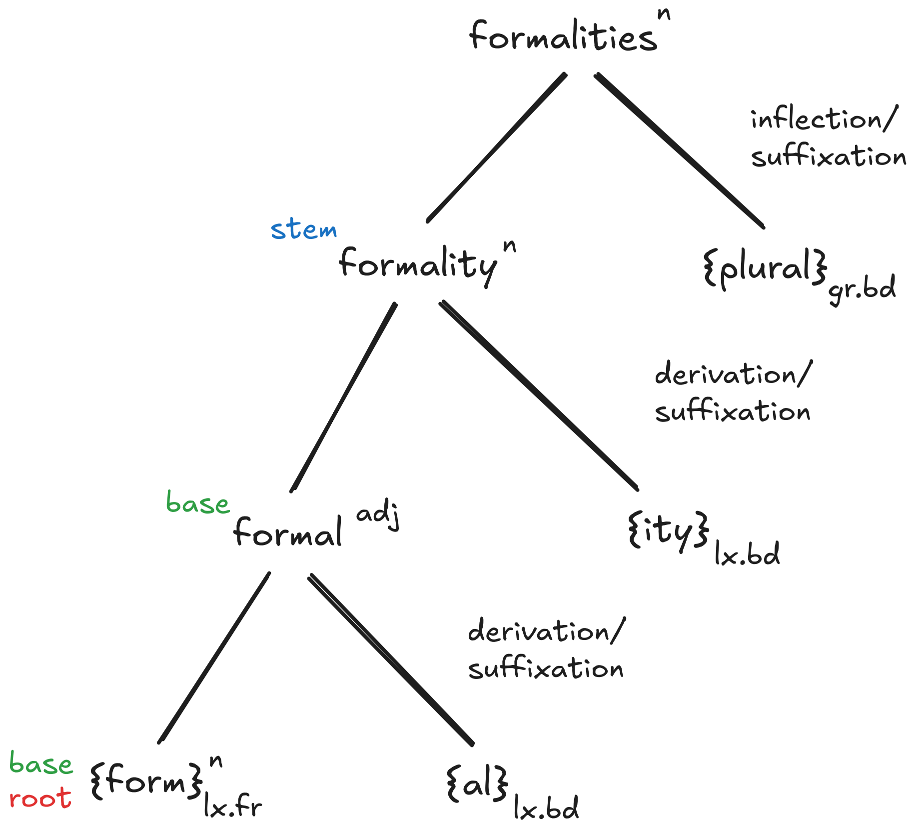
(Kortmann 2020: 54)



removals n
inflection
removal n
stem
{plural} gr.bd
derivation/
suffixation
{remove} v lx.fr
{al} lx.bd
base
root

Exercise: determine base, root, stem

Determine the base, root, and stem of the word *formalities*.



Allomorphs

Complementary distribution

Complementary distribution: The regularity underlying the complementary distribution of these three allomorphs can be described as in (2):

(2) /-ɪz/: after sibilants (or: hissing sounds), i. e. /s/, /z/, /ʃ/, /ʒ/, /ʧ/, /ʤ/
/-s/: after all other voiceless consonants, i. e. not for /s/, /ʃ/, /ʧ/
/-z/: after all other voiced consonants and all vowels (except for /z/, /ʒ/, /ʤ/)

Kortmann (2020): 55

Plural allomorphs

{D}	**‘past tense’**	**examples**
/t/	after voiceless consonants except /t/	*worked, picked*
/d/	after vowels after voiced consonants except /d/	*allowed* *seemed, loved, used, scribbled*
/ɪd/	after /t, d/	*insisted, decided*

Herbst (2010): 85

Past tense allomorphs

{S}	**'plural'**	
/s/	after voiceless consonants except /s, ʃ, tʃ/	*critics, shapes, moments, elements, discs*
/z/	after vowels after voiced consonants except /z, ʒ, dʒ/	*contours, shoes* *paintings, rhythms, soles, compositions, coastlines*
/ɪz/	after /s, z, ʃ, ʒ, tʃ, dʒ/	*references*

Herbst (2010): 90

Overview of allomorphy in English

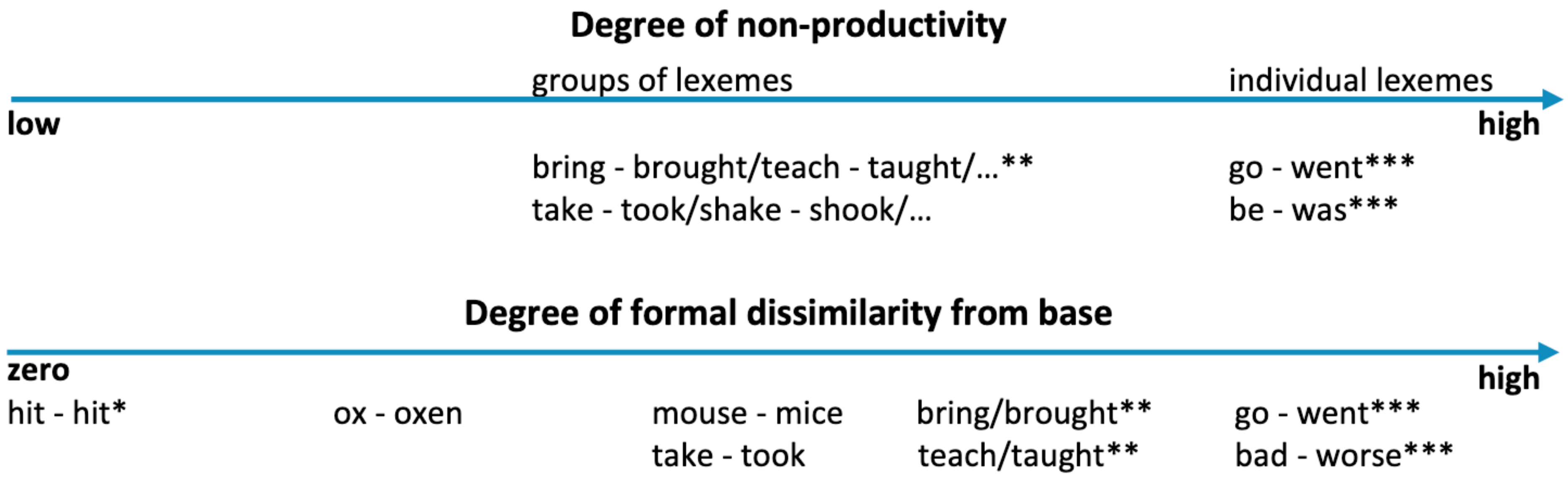
Tab. 3.5: Overview of the allomorphs of the inflectional morphemes in present-day English

morpheme	phonologically conditioned			morphologically conditioned	
	form	environment; stem ending in	examples	form	examples
{plural}	/z/ /s/ /ɪz/	vowel or voiced consonant[1] unvoiced consonant[1] sibilant (/z/,/s/, /ʒ/, /ʃ/, /dʒ/, /tʃ/)	*cars*, *dogs*, *worries* *cats*, *ducks*, *rips* *houses*, *matches*	Ø i-mutation (*umlaut*) /ən/	*sheep*, *aircraft*, *fish* *men*, *mice*, *geese* *oxen*
{genitive}	/z/ /s/ /ɪz/ Ø	vowel or voiced consonant[1] unvoiced consonant[1] sibilant (/z/,/s/, /ʒ/, /ʃ/, /dʒ/, /tʃ/) after /z/ (and gen. plural in general)	*dog's*, *John's* *cat's*, *Pat's* *George's*, *Jones's* *Dickens'*, *the boys'*	Ø (polysyllabic Greek names ending in /z/)	*Socrates'*, *Xerxes'*
{3rd pers.}	same as {plural}			suppletion	*is*
{ing}	no allomorphs				
{ed$_1$}, {ed$_2$}	/d/ /t/ /ɪd/	vowel or voiced consonant[2] unvoiced consonant[2] dental (/d/, /t/)	*loved*, *remembered* *kissed*, *stopped* *founded*, *rented*	Ø gradation (*ablaut*) /ən/ (often with gradation) portmanteau forms suppletion	*put*, *let*, *hit*, *cut* *sang/sung*, *began/begun* *taken*, *spoken*, *been* *did*, *taught*, *told*, *stood* *was*, *were*, *went*
{er}	no phonologically conditioned allomorphs			suppletion	*better*, *worse*
{est}	no phonologically conditioned allomorphs			suppletion	*best*, *worst*
{th}	no phonologically conditioned allomorphs			suppletion	*first*, *second*, *third*

Explanatory notes: 1) except voiced (/z/,/ʒ/, /dʒ/) and voiceless sibilants (/s/, /ʃ/, /tʃ/) respectively
2) except voiced (/d/) and voiceless dental (/t/) respectively

Schmid (2016): 61

Irregular allomorphs



Kortmann (2020): 58

Phonological conditioning of allomorphs

Types

- “**affix** allomorphy, where a particular free morpheme, i. e. base, demands the choice of a particular (form of a) bound morpheme,
- and **base** allomorphy, where, vice versa, a particular affix (be it derivational or inflectional) demands the choice of a particular form of the base.”

(Kortmann 2020: 58)

Examples

(6)	electric	/ɪ’lektrɪk/	electric-ity	/ɪˌlek’trɪs-/ + /-ɪti/
	electric	/ɪ’lektrɪk/	electric-ian	/ɪˌlek’trɪʃ-/ + /-(ə)n/
	invade	/ɪn’veɪd/	invas-ive	/ɪn’veɪs-/ + /- ɪv/
	invade	/ɪn’veɪd/	invas-ion	/ɪn’veɪʒ-/ + /-(ə)n/
	part	/pɑːt/	part-ial	/pɑːʃ-/ + /-(ə)l/
	infuse	/ɪn’fjuːz/	infus-ion	/ɪn’fjuːʒ-/ + /-(ə)n/
	convulse	/kən’vʌls/	convuls-ion	/kən’vʌlʃ-/ + /-(ə)n/
(7)	angel	/’eɪndʒəl/	angel-ic	/æn’dʒel-/ + /-ɪk/
	miracle	/’mɪrək(ə)l/	miracul-ous	/mɪ’ræk-/ + /-jʊləs/
	particle	/’pɑːtɪkəl/	partic-ular	/pə’tɪk-/ + /-jʊlə/
	photograph	/’fəʊtəˌgrɑːf/	photograph-ic	/ˌfəʊtə’græf-/ + /-ɪk/

Kortmann (2020): 59

(6): Consonant change of the final sound of the base due to **coarticulation**.
(7): Vowel changes of the base due to **stress shift** triggered by derivation.

Exercise: analyse plural allomorphs

In which way do the plural forms of the following nouns differ?

cat, *horse*, *woman*, *mouse*, *dog*, *book*, *fish*, *orange*, *apple*, *car*, *child*, *tax*, *ox*, *cliff*, *deer*

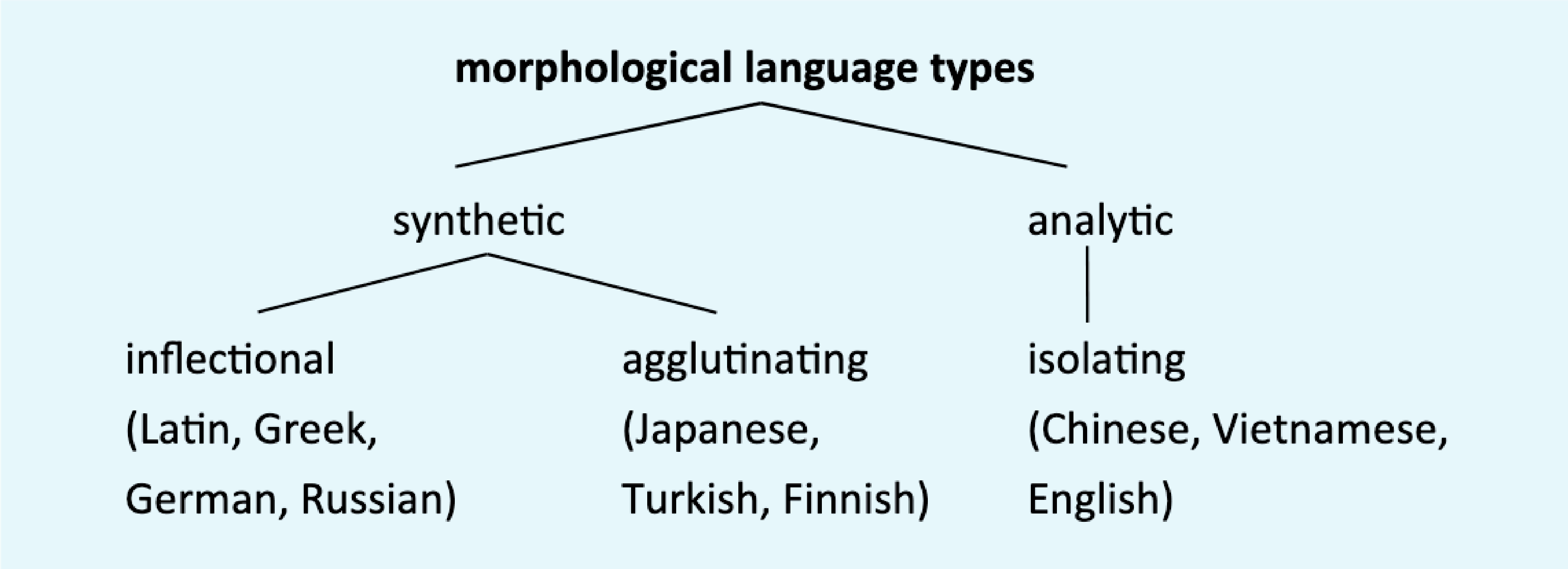
Options:

1. `/s/`
2. `/z/`
3. `/ɪz/`
4. irregular

Solution

1. /s/	2. /z/	3. /ɪz/	4. irregular
cats	*dogs*	*horses*	*women*
books	*apples*	*oranges*	*fish*
cliffs	*cars*	*taxes*	*children*
			oxen
			deer
			mice

Morphological language types



Kortmann (2020): 85

Morphology Review Quiz (Interactive)

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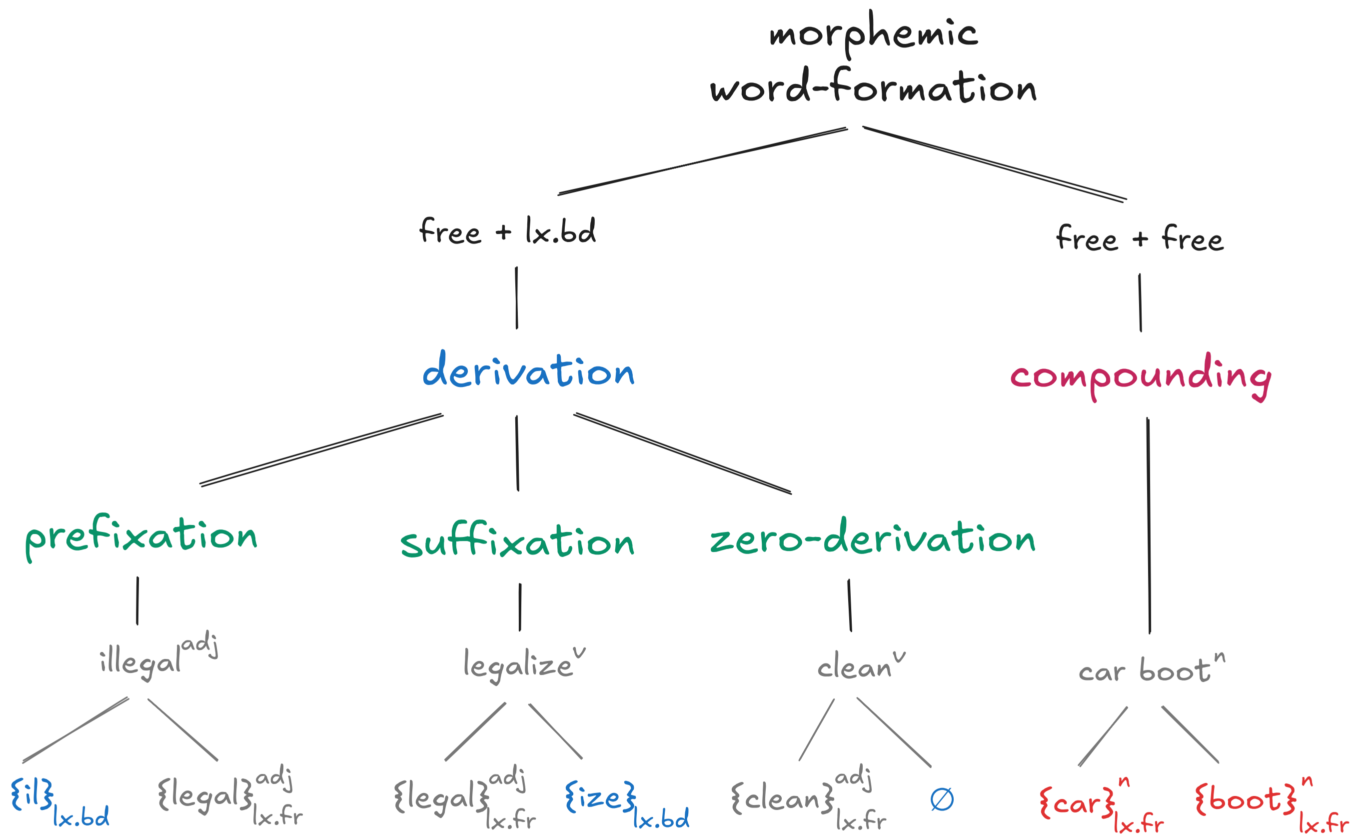
Word-formation

Morphemic word-formation



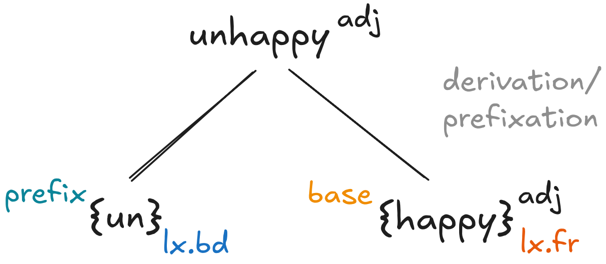
Derivation

Derivation vs inflection

Kortmann (2020): 86

inflection	**derivation**
part of the grammar	part of the lexicon
produces word forms (by means of suffixation)	produces lexemes (by means of prefixation or suffixation)
never changes word class	can change the word class
usually fully productive within one word class (e. g. possessive -*s* for all nouns)	only productive for subgroups of word-classes (e. g. -*ity* versus -*dom*)
very small inventory of inflectional morphemes with few very general meanings	large inventory with many relatively specific meanings
the meaning of the word form is predictable (e. g. *boys*, *walked*, *higher*)	the meaning of a new lexeme is not always predictable (*singer* = 'somebody who sings', but *sweater* = 'somebody who sweats'?)
closed class (two possible candidates for additional inflectional morphemes: negation and adverb-forming -ly)	more open class (e. g. -*hood*, -*dom*, -*(a)holic* in *workaholic*, *chocaholic*, *shopaholic*)
further away from the root (only after the derivational suffixes)	closer to the root
strongly syntactically determined	hardly syntactically determined

Prefixation



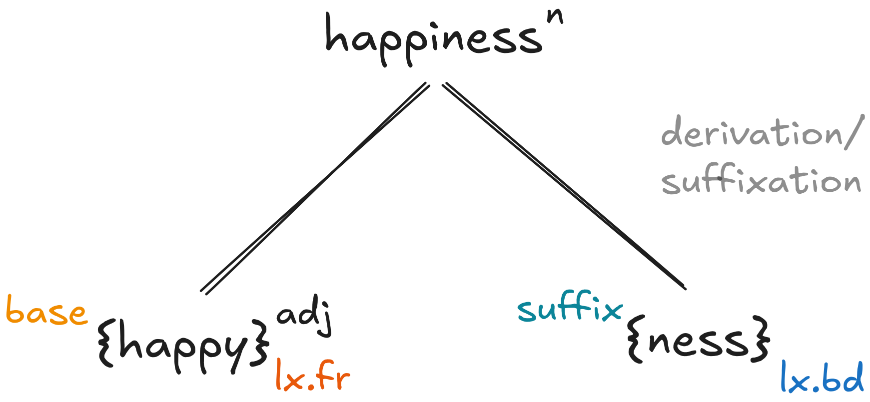
- **formal structure**
  - → prefix: $\{A\}_{\text{lx.bd}}$
  - → base: $\{B\}_{\text{lx.fr}}$
- **function**
  - → modification of meaning
  - → usually word-class preserving (exceptions: *asleep*, *enlarge*)

Morphemes

semantic type	prefix
negative	*a-*
	dis-
	in- (il-, ir-, im-)
	non-
	un-

semantic type	prefix
time and order	*ex-*
	fore-
	post-
	pre-
	re-
number	*bi-, di-*
	poly-, multi-
	semi-, demi-
	tri-
	uni-, mono-

Suffixation



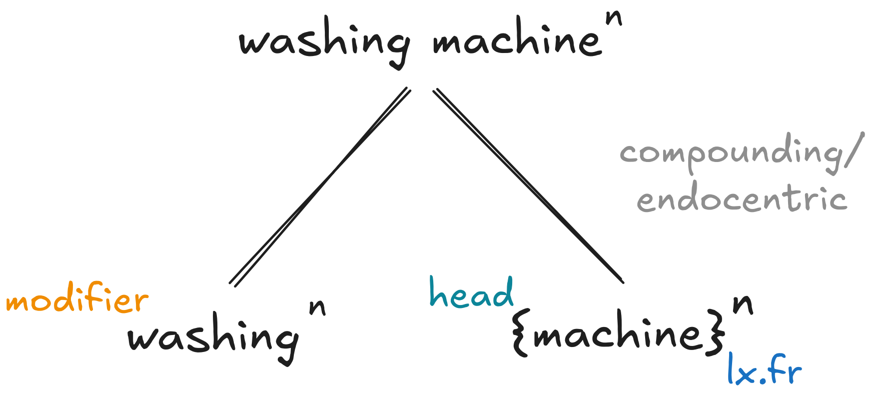
- **formal structure**
  - → base: $\{A\}_{lx.fr}$
  - → suffix: $\{B\}_{lx.bf}$
- **function**
  - → (abstract) modification in meaning (e.g. concept type)
  - → often word-class changing (exceptions: *greenish*, *kingdom*)

Examples

	> noun	> adjective	> verb
noun		*part-ial*	*woman-ize*
		partic-ular	*class-ify*
		fashion-able	*haste-n*
		boy-ish	*orchestr-ate*
		clue-less	
adjective	*electric-ity*		*modern-ize*
	tough-ness		*pur-ify*
	free-dom		*activ-ate*
	social-ist		*black-en*
	warm-th		
verb	*infus-ion*	*invas-ive*	
	pronunc-iation	*hope-ful*	
	develop-ment	*hope-less*	
	employ-er	*drink-able*	
	employ-ee	*frighten-ing*	

Kortmann (2020): 61

Compounding



Endocentric compounds

- **semantic structure**: 'a certain type of machine: for washing'
- **head**: *machine*
  - → determines word class: *machine*^{n} > *washing machine*^{n}
  - → stem for inflectional suffixes: *washing machines*, not **washings machine*
- **modifier**: *washing*
  - → qualifies head semantically
  - → often carries primary stress

Examples

	noun	**adjective**	**verb***
noun	(13)	*waterproof*	*proofread*
		knee-deep	*gatecrash*
	PROTOTYPE	*sky-high*	*babysit*
		airsick	*tailor-fit*
		scandal-weary	*day-dream*
adjective	*small talk*	*deaf mute*	*fine-tune*
	deadline	*ready-made*	*double-book*
	wild card	*far-fetched*	*free associate*
	greenhouse	*short-sighted*	*short-list*
verb	*talk show*	*fail-safe*	*sleepwalk*
	playboy	*?*	*freeze-dry*
	cry-baby		
	cutthroat		
	pickpocket		

Kortmann (2020): 64

Semantic types

- **endocentric** compounds
  - → **structure**: A + B denotes a special kind of B
  - → **examples**: *darkroom*, *small talk*
- **exocentric** compounds
  - → **structure**: A + B denotes a special kind of an unexpressed semantic head (e. g. ‘person’ in the case of *skinhead* or *paleface*)
  - → these compounds often have metonymic character, when one part stands for the whole (e. g. *paleface* for ‘a person with a pale face’)
  - → **examples**: *egghead*, *blockhead*, *blackhead*, *birdbrain*, *redneck*, *greenback*, *paperback*
- **appositional** compounds
  - → **structure**: A and B provide different descriptions for the same referent
  - → **examples**: *actor-director*, *actor-manager*, *writer-director*, *poet-translator*, *maidservant*
- **copulative** compounds
  - → **structure**: A + B denotes ‘the sum’ of what A and B denote, i. e. the denotation of a copulative compound would be incomplete with one of the two elements A and B missing
  - → **examples**: *spacetime*, *tractor-trailer*, *deaf mute*, *sleepwalk*, *freeze-dry*

(Kortmann 2020: 65)

Exercise

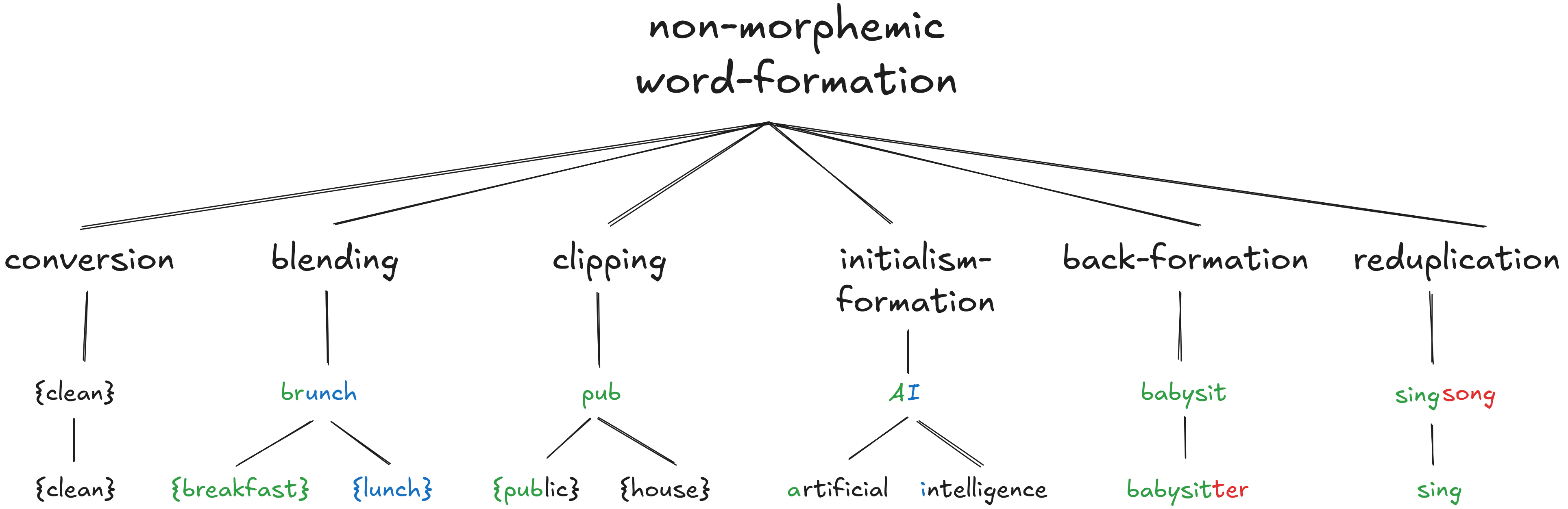
Use the following terms to describe the meaning relationship between the component parts of the compounds below:

> endocentric — exocentric — appositional — copulative compound

1. *greenhouse*: endocentric
2. *knee-deep*: endocentric
3. *pot head*: exocentric
4. *Alsace-Lorraine*: copulative

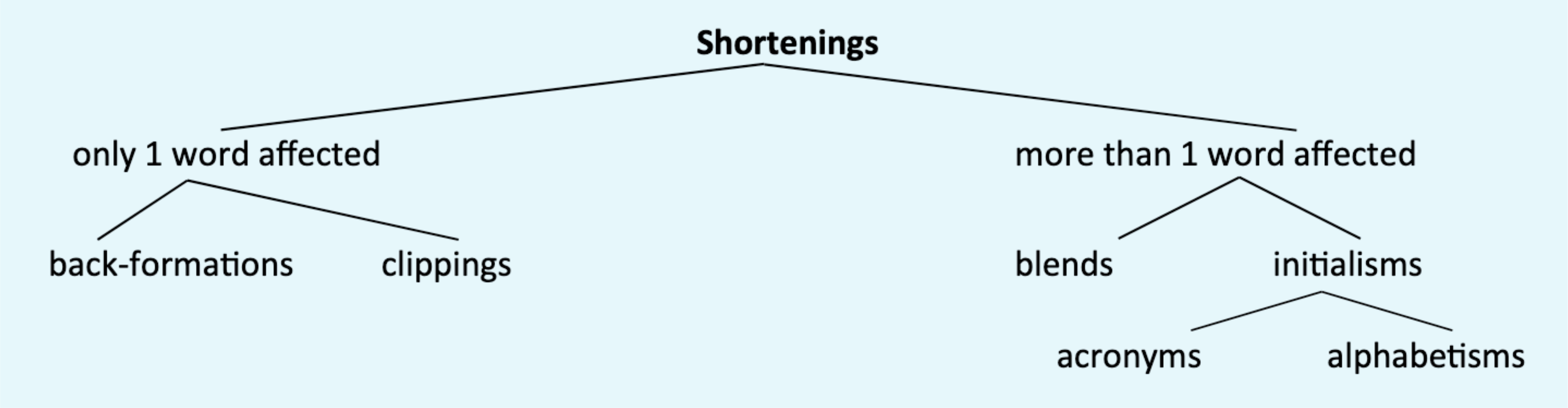
Non-morphemic word-formation

Overview



Process	Features	Examples
Clipping	parts of words are deleted without a change in meaning	*bike*, *exam*, *fridge*
Blend	forms and meanings of words are merged	*smog*, *brunch*
Acronym	shortened form retaining the initial letters of compounds or other fixed sequences of words; pronounced as words	*NATO, laser, AIDS*
Alphabetism	shortened form retaining the initial letters of compounds or other fixed sequences of words; pronounced as sequences of letters	*CNN*, *OED*, *USA*
Back-formation	word-class-changing word-formation process which deletes a morpheme or morpheme-like element; longer form pre-existing	*babysitter*^{n} > *babysit*^{v}
Reduplication	rare word-formation process repeating a word or word-like element either identically or in a slightly varied form	*walkie-talkie*, *hip-hop*

Shortenings



Kortmann (2020): 72

Differences between morphemic and non-morphemic word-formation

Feature	Morphemic WF	Non-morphemic WF	Examples
Building blocks	morphemes	parts of words	*br-unch*
Combination	adding morphemes	removing parts	in-*flu-enza*
Serialization	concatenative	interspersed	*CHuckLE x snORT*
Regularity	clear rules, predictable patterns	heterogeneous, high variability	*Br-angelina*
Cognitive processing	largely unconscious	conscious coining	*BBC*

Productivity

The degree to which a certain word-formation process is being employed to form new words.

Examples

Still productive: suffixation with **{ness}**:

- {kind}{ness}
- {happy}{ness}
- {brave}{ness}
- ...

No longer productive: suffixation with **{th}**:

- {strong}{th} > strength
- {long}{th} > length
- {wide}{th} > width
- ...

Transparency

The degree to which there is a clear (transparent) mapping between the form and function/meaning of linguistic units.

Examples

transparent:

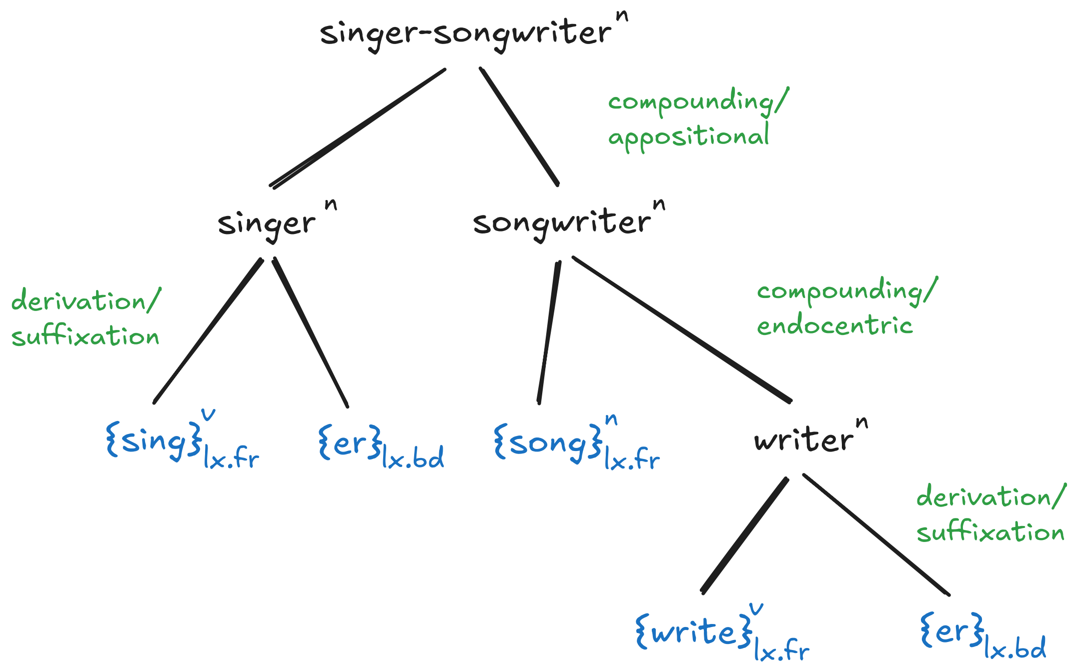
- *player*: 'somebody who plays'
- *wine bottle*: 'a bottle of wine'

intransparent/opaque:

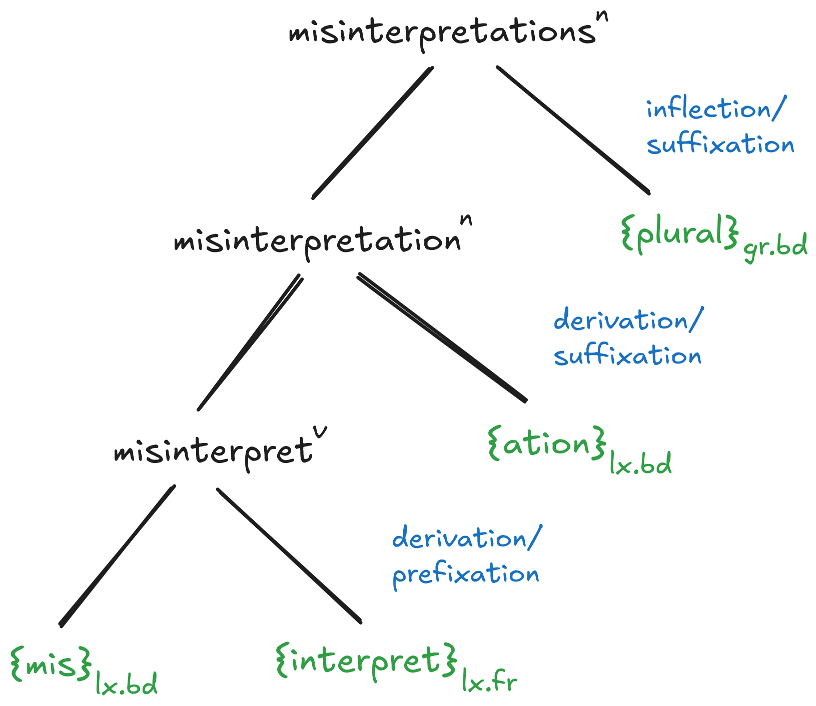
- *caretaker*: *'somebody who takes care'
- *undertaker*: *'somebody who undertakes sth.'
- *to kick the bucket*: *'to strike a bucket with your foot'

Morphological analysis

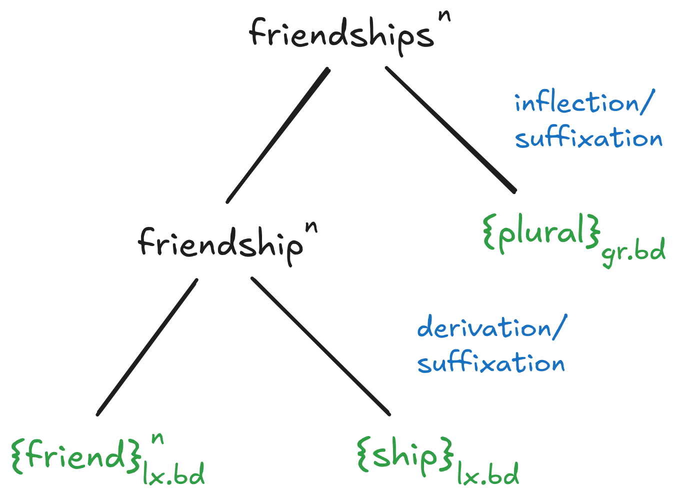
singer-songwriter



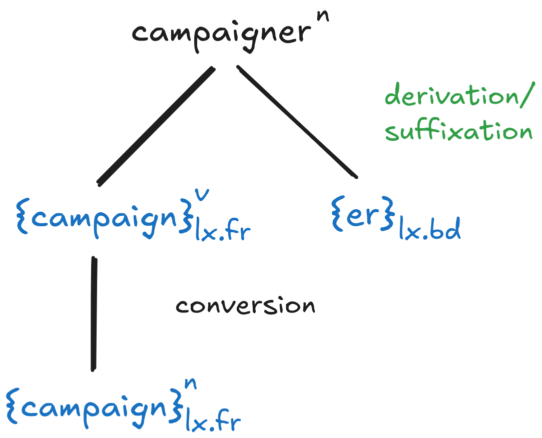
misinterpretations



friendships



campaigner


campaigner^{n}
derivation/
suffixation
{campaign}$^{v}_{lx.fr}$
{er}$_{lx.bd}$
conversion
{campaign}$^{n}_{lx.fr}$

Determine the word-formation process I

Analyse the word-formation processes involved in the following complex words.

1. *childhood*: derivation / suffixation
2. *paperback*: compounding / exocentric
3. *study-bedroom*: compounding / copulative
4. *foreigner*: derivation / suffixation
5. *paleface*: compounding / exocentric
6. *Oxbridge*: blending
7. *CGEL*: initialism

Exercise: determine the word-formation process II

By means of which **word-formation process** do we arrive at the following lexemes?[1]

1. *to enthuse*: **back-formation** from *enthusiasm*: N>V
2. *laser*: **acronymisation** from *Light Amplification by Stimulated Emission of Radiation*
3. *lab*: **clipping** from *laboratory* N>N
4. *Hinglish*: **blending** of *Hindi*[n] and *English*[n]
5. *BBC*: **alphabetism** of *British Broadcasting Corporation*
6. *sext*: **blending** of *sex*[n] and *text*[n]
7. *to hoover*: **(N) coinage** after American industrialist H. W. Hoover (1849–1932), then **conversion** N>V

1. When two or more processes are involved specify their order.

References

Herbst, Thomas. 2010. *English Linguistics: A Coursebook for Students of English*. Berlin: De Gruyter Mouton.
Kortmann, Bernd. 2020. *English Linguistics: Essentials*. J.B. Metzler.
Schmid, Hans-Jörg. 2016. *English Morphology and Word-Formation - An Introduction*. 2nd ed. Berlin: Erich Schmidt Verlag.